

NEXTorr[®] Z pumps

Operating instructions

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Provisional

making innovation happen, together



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SAES Getters S.p.A.
viale Italia 77
20045 Lainate (MI), Italy

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General information

Validity

This equipment is intended for use by professionals only, operating in a proper laboratory environment.

These operating instructions give important information about the functioning of NEXTorr® Z pumps. Following these instructions guarantees a safe use of the equipment, according to the applicable EU guidelines and regulations. More detailed information about the technical specifications of NEXTorr Z pumps is given in the *Technical specifications* document.

NEXTorr Z pumps are intended to be operated through SAES power supplies only. The safety specifications given in the following are not guaranteed in case the pump is operated using different controllers.

SAES disclaims all liability for any event caused by improper use by untrained personnel and/or full or partial non-compliance with these instructions.

SAES reserves its right to partially or entirely amend this manual in future editions without prior notification.

Reproduction or translation, even partial, of this manual is forbidden without prior and express consent from SAES.

Conventions

Safety instructions

This manual conforms to the following classification to identify possible hazards for users and instrumentation and to give important information to the reader.

Warning!

A warning message puts in evidence an action or procedure that could lead to personal injury or death.

Caution!

A caution message draws the attention to an action or procedure that could cause damage to the equipment.

Note

Notes give important information that is worth highlighting.

Pictographs

The following pictographs are used in this document, in compliance with EN ISO 7010:2012/A7:2017.



Read the user manual before operating the equipment.



Wear foot protection.



Attention! Carefully read this important information.



Warning! Risk of electric shock.



Warning! Risk of high temperature.



Warning of sharp objects.



Warning! Risk of hand injury.



Warning! Flammable solid.



Warning! High magnetic field.



Pacemaker and implanted cardioverter-defibrillator holders should keep a sufficient safe distance.

Product description

NEXTorr[®] Z pumps are a family of vacuum combination pumps consisting of a non-evaporable getter (NEG) cartridge and a compact sputter-ion pump (ION), which are connected to the same ConFlat[®] (CF) flange.

The NEG element is made of sintered ZAO[®]1 (Ti-Zr-V-Al) porous disks, characterised by an extremely-low particle release; it is able to efficiently pump all chemically-reactive gases (H₂, H₂O, N₂, O₂, CO, CO₂), while the sputter-ion pump provides a sufficient pumping for non-getterable species, such as CH₄ and noble gases.

NEXTorr Z pumps are designed and engineered to work in a variety of vacuum systems where stringent ultra-high vacuum and extremely-high vacuum (UHV-XHV) conditions should be met, such as particle accelerators, scanning electron microscopes, portable vacuum suitcases and semiconductor inspection tools.

A schematic view of a NEXTorr Z pump is given in Figure 1. Its main components are:

- a. the NEG cartridge,
- b. the sputter-ion pump,
- c. the NEG connector,
- d. the ION connector,
- e. the connecting CF flange.

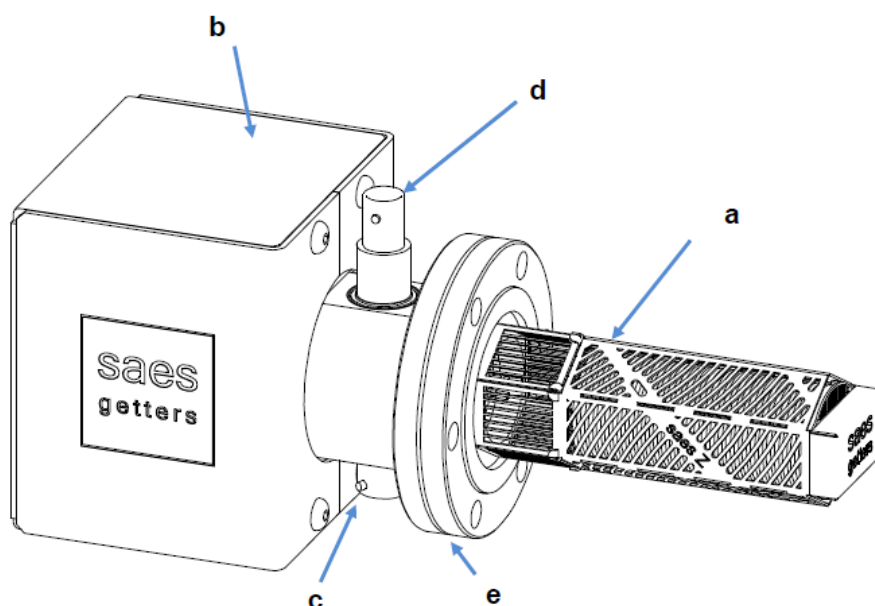


Figure 1 – Schematic view of a NEXTorr Z pump.

The NEG cartridge consists of an AISI 316 L stainless-steel screen containing sintered porous ZAO[®]1 getter disks, which are positioned in one or more parallel stacks, depending on the model. The screen protects the getter disks from any accidental contact, contamination or damage. Composition, porosity and surface area of the getter disks, as well as their arrangement inside the cartridge, are optimised to maximise both pumping speed and sorption capacity.

A heating element is needed to activate the getter disks. Each stack of disks has a heating element passing through it and consisting of a tantalum wire inside an alumina insulator. The getter cartridge and its heating element are integrated into a CF flange, which is used to install the pump on the vacuum system. Several flange sizes are available, depending on the pump model and on the dimensions of the getter cartridge. Base flanges of NEXTorr Z pumps are made of austenitic stainless steel. All of them are standard ConFlat[®] (CF) and He leak tested down to $1 \cdot 10^{-10}$ mbar·l/s.

A sputter-ion pump ("ION" in the following) with a permanent magnetic yoke is connected to the opposite side of the same flange through a short neck, on which both NEG and ION connectors are welded. This particular design enhances the synergy between the getter cartridge and the sputter-ion pump in terms of:

- minimal conductance limitation at the ION inlet;
- reduced backstreaming (gases occasionally released from the ION element can be intercepted by the getter cartridge);
- increased pumping of CH₄ and other hydrocarbons;
- reduced titanium fine dust release from the ION element.



Warning!

High magnetic field! Implanted pacemakers and cardioverter-defibrillators may be affected. Keep a sufficient safe distance.



Note

Do not put any electronic device in close proximity of ION magnets, to avoid possible malfunctioning caused by their magnetic field.

Transport and storage

Each pump is vacuum sealed with a blank-off flange welded to a crimped pinch-off copper tube. The pump is further vacuum sealed inside a static-shielding moisture-barrier bag. The shipping package envelops the pump in moulded polyurethane foam.

Store the pump indoors only, in a dry and proper location. Use its original blank-off flange to protect it. Avoid removing the magnets from the pump, as this may result in a loss of stability of the stored equipment.



Caution!

Special care must be exercised when handling the pump. Accidental dropping or mechanical shocks may permanently damage the getter disks and/or the heating elements.



Note

Do not break the vacuum sealing until the final installation of the pump in the vacuum system. Prolonged exposure to the environment can be detrimental to the pump, due to the presence of moisture, dust and other contaminants.

Installation

Remove the pump from the package and the moisture-barrier bag and verify the integrity of the vacuum-sealed blank-off flange.



Warning!

Wear foot protection while carrying and installing NEXTorr Z pumps.

Unscrew the bolts and nuts and remove the blank-off flange without damaging the NEG cartridge inside (cf. Figure 2). Keep the blank-off flange and reuse it, should the pump be removed from the vacuum system and/or shipped elsewhere or back to SAES for inspection.



Warning!

Do not remove the protective plastic cover from the copper pinch-off tube connected to the blank-off cover. The pinch-off tube has sharp edges, which may cause injuries.



Caution!

Removing the plastic cover may damage the pinch-off tightness and cause an unwanted venting of the pump to the atmosphere.

Install the NEXTorr Z pump on the vacuum system, connecting its base flange to an available port with the same flange size. Use a new copper gasket and appropriate bolts and nuts. The pump can operate in any orientation.

When primary pumping is restarted, leak check the vacuum system to verify that the pump is properly connected and the system is tightly sealed.



Note

As for all UHV equipment, good standard practices must be followed to ensure a correct functioning of the pump and to avoid any contamination on its components that will be exposed to vacuum.

Do not use bare hands and always wear clean, lint-free gloves to handle the pump and operate on a clean, dust-free desk.



Caution!

Avoid any contamination of the NEXTorr Z pump with oils of primary mechanical pumps. Avoid contamination by sulphur, fluorine and chlorine compounds too.

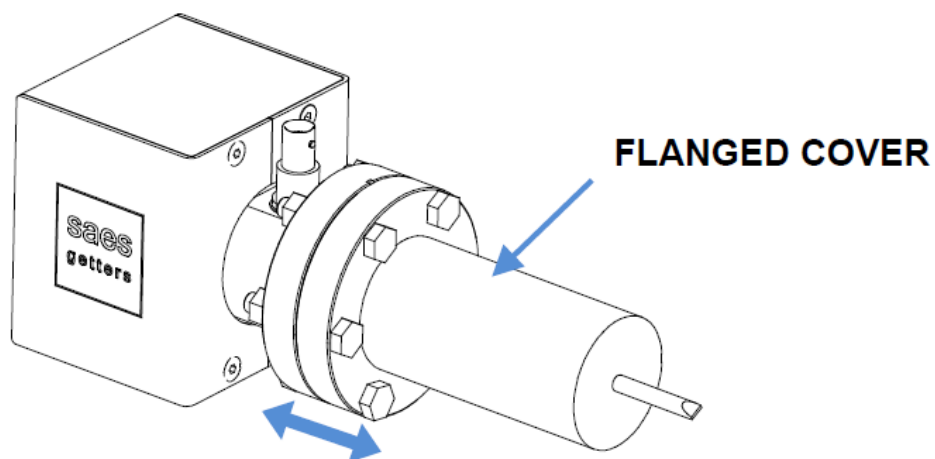


Figure 2 – Blank-off flange removal.

Use

NEXTorr Z pumps should be operated indoor, in a proper laboratory environment, by professional and trained users only. They are not intended for use in potentially-explosive atmospheres or environments.

Environmental conditions	
Ambient temperature	0–40 °C
Elevation	Max. 2000 m a.s.l.
Relative humidity	Max. 80%
Pollution degree	2
Wet location	Not applicable

The following provides essential information about baking the pump and operating the NEG and ION elements; for further details, consult the *Technical specifications* document.



Note

Carefully read the operating instructions and any other document provided by SAES before starting to operate the pump and its controllers.

Bake-out

When a UHV system undergoes a bake-out, it is in general advisable to bake also the ION element of NEXTorr, in order to enhance the degassing from its inner surface area and to lower the final base pressure.

Standard heating tapes or other heating means can be used to bake the ION element. During the bake-out, the ION element must be switched off.

When magnets and red covers are installed, the maximum bake-out temperature of a NEXTorr Z pump is **150 °C**. If the system is going to be baked at higher temperature, magnets and red covers should be removed. In any case, the maximum allowable baking temperature for the pump body without magnets is **250 °C**. Cables should be removed if the bake-out exceeds their maximum allowed temperature.



Warning!

Beware of high temperature! Do not touch the pump while it is hot.

To remove the magnets, follow these steps (cf. Figure 3):

1. Unscrew the four small screws;
2. Remove the back cover and the two side covers;
3. Pull out the magnetic yoke.

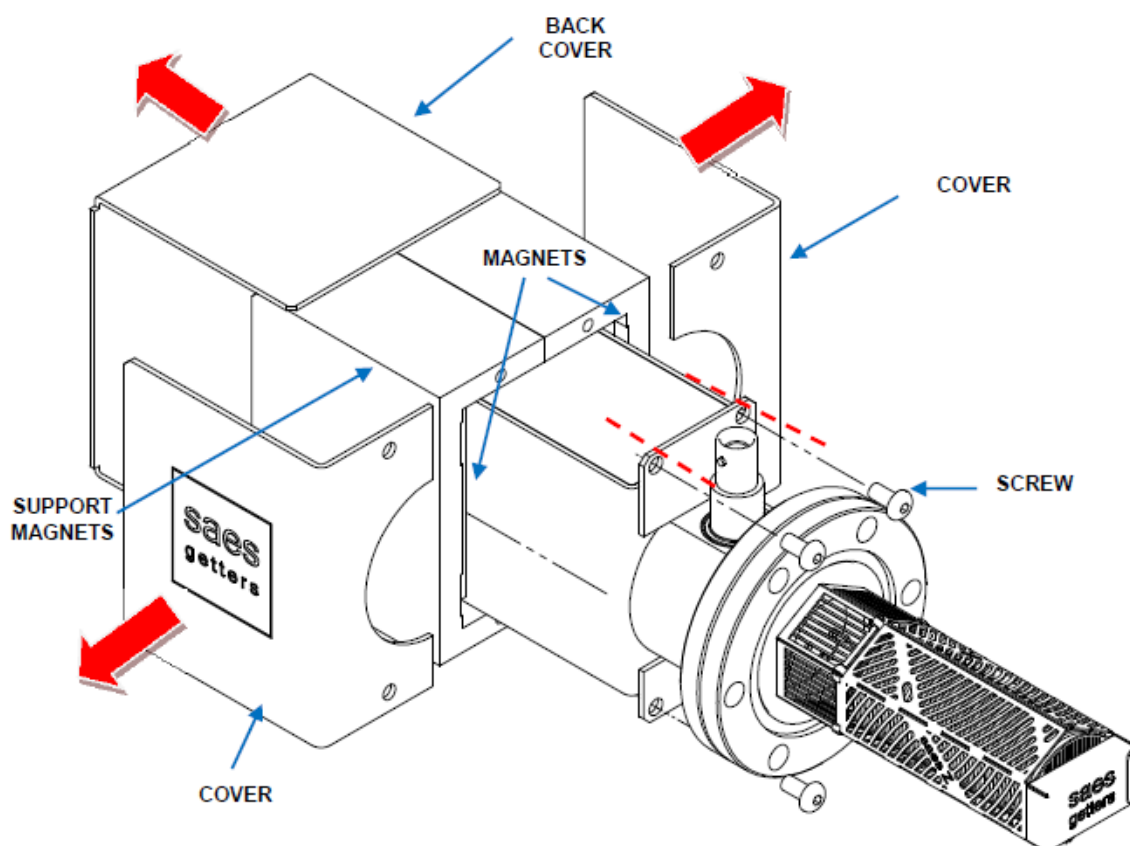


Figure 3 – Removal of red covers and magnets from a NEXTorr Z.



Warning!

To avoid trapping of fingers when handling strong magnets, do not try to separate the two halves of the magnetic yoke.

Powering the pump

Do not try to operate the pump in air; always verify that the pump is properly installed on a leak-tight vacuum system before connecting the cables and switching on the controllers.

SAES recommends using its dedicated cables and controllers to power NEXTorr Z pumps. Standard cables are 3 m long; longer cables, as well as radiation and/or high-temperature resistant ones, are available on request.

Please refer to the *Technical specifications* document or ask your local SAES representative for further details about controllers and cables available for each NEXTorr Z model.

**Warning!**

To avoid any risk of electric shock, the vacuum system housing the pump should be properly grounded. Connecting the pump to a grounded power supply is not sufficient to ensure safe operating conditions for the users.

**Caution!**

The maximum applicable current for the NEG element is 5 A. Higher currents may damage the pump and the connector.

**Note**

Safe operation of NEXTorr Z pumps is guaranteed using SAES controllers and cables only.

ION operation

Always check the pressure inside the vacuum system before switching on the ION element: if the pressure is lower than 10^{-5} Torr, the ION element can be operated.

Switch on and off the ION element a couple of times before NEG activation, to avoid any partial saturation of the NEG cartridge due to gas contamination generated by plasma ignition inside the ION element.

The ION element must be switched off during NEG activation, to avoid the sorption of undesired H₂ loads desorbed by the NEG cartridge while it is hot.

If the ION element is working properly, its power supply will display a current reading, indicating that the plasma discharge has been ignited. The current drawn by the ION element should be proportional to pressure; cf. the *Technical specifications* document for further details.



Note

Refer to the power-supply operating instructions for further information about the operation of the ION element.

NEG activation, operation, and conditioning

A new NEG cartridge should be activated in order to develop its pumping capability. Activation is achieved by heating the getter disks under vacuum with the heater up to a suitable temperature for a sufficient length of time.

Although there are no strict rules, as initial conditions may drastically vary from one system to another, NEG activation of a NEXTorr Z pump should preferably start when pressure is lower than **10⁻⁴ Torr**.

Typically, the activation process lasts **1 h** and the activation temperature is **550 °C**.

The *Technical specifications* document provides the default activation parameters (in terms of supplied voltage and power) of each NEXTorr Z model, assuming that the NEG cartridge is not confined in a tight tube or volume. As a matter of fact, the power needed to keep the activation temperature also depends on the installation configuration: when a pump is installed in a pipe or in a small-size chamber, the chamber walls act as a thermal shield and a lower power is sufficient to keep the NEG disks at the same activation temperature.

No power is needed once activation is completed, since the NEG element of a NEXTorr Z is designed to operate at room temperature in standard UHV/XHV environments.

In this pressure range, a single sorption cycle can be quite long and even last for several years, depending on the residual gas composition and on the gas sources inside the vacuum system. More details about the typical pumping speed and sorption capacity of available NEXTorr Z models are given in the *Technical specifications* document.

A freshly-activated NEG can provide its full pumping speed, which then progressively decreases from this initial value as surface saturation progresses. When the pumping speed falls below a minimum acceptable threshold, the pump should be reactivated to recover its initial pumping performance and to start a new sorption cycle. Reactivation is performed by again heating the NEG element to 550 °C and keeping it there for 1 h; it is also required every time the pump is vented to air.

The entire capacity of the getter material will be over after several successive sorption cycles only. In case of standard operation at room temperature in the UHV/XHV range, a NEXTorr Z can be reactivated more than a hundred times overall.

A further operating mode is the so-called *conditioning*, when a power lower than the one necessary for activation is supplied to the NEG element. Under these conditions, the getter disks are kept at about **200–300 °C** and a significant fraction of the surface adsorption sites is continuously cleared by thermal diffusion. As a result, pumping speed remains high for a prolonged time. Keeping the NEG element in conditioning mode may be useful, for instance, to reduce bake-out duration or to improve the base pressure during a pump-down.



Warning!

Beware of high temperature! During NEG activation and conditioning, the piping where the cartridge is installed may be hot.

Venting

Do not vent the system when the NEG temperature is above **80 °C**. In any case, it is recommended to vent when the getter is at room temperature, in order to reduce the surface oxidation and to ensure a prolonged pump's lifetime. After turning off the heater, it takes at least **90 min** to cool the getter down to 80 °C. This time may be longer depending on the distance from other surfaces and on their temperature.

A controlled dry N₂ venting is an effective way to preserve the pump performance in case of several venting/reactivation cycles. A further improvement is the use of pure argon or neon as protective gas during maintenance operations on the vacuum system.

Although the ZAO alloy is an improved and robust getter material, it is a good practice to avoid any sudden air venting when the getter pump is hot, *i.e.*, during activation, conditioning at 200 °C or bake-out.

When the getter is hot, a sudden air venting due to a vacuum failure may lead to an uncontrolled exothermic reaction; the resulting temperature increase and the final oxidation of the getter disks depend on the system conditions and geometry.

In any case, should an accidental air venting occur while the getter is being heated:

- Switch off immediately the power supply of the NEG element, as well as any other additional or potential heating source, and try to recover a suitable vacuum level.
- Do not touch or try to remove the pump until the getter cartridge cools down.

**Warning!**

ZAO-based getters are classified as flammable solids according to EC Regulation 1272/2008. Please refer to their *Safety Data Sheet* for further information about packaging, handling, disposal, and possible hazards.

Maintenance

The NEG cartridge, ION element and connecting CF flange are, by design, one single device. Replacement of the NEG cartridge is therefore not possible. At the end of its working life, when either the NEG or the ION element is exhausted, the whole NEXTorr Z pump must be replaced.



Warning!

Disconnect the pump from its power supplies before carrying out any operation on it.

Be sure that the power supply units are switched off before unplugging the high-voltage cable connectors. Wait at least 10 s before removing the connectors from their respective power supplies.



Caution!

Do not clean the pump or its component with any kind of detergent or solvent. Avoid in particular sulphur, fluorine and chlorine compounds, as they would permanently damage the getter.

Service

Please contact SAES Customer Service or your local sales representative for return requests.



Note

Please note that SAES cannot accept any pump which contains biological or chemical hazards or radioactive substances. Please clearly inform SAES Customer Service should this have happened during pump use, in order to discuss adequate solutions.

Sales & service locations

Europe, Middle East and Africa

SAES Getters S.p.A.
Viale Italia 77
20020 Lainate (MI), Italy

Tel: +39 02 93178 1 – Fax: +39 02 93178 320

European Customer relations

Tel: +39 02 93178 402 – Fax: +39 02 93178 320

E-mail: CRM_SALES@saes-group.com

Asia and Oceania

SAES Getters S.p.A. - Japan Technical Service Branch Office

2nd Gotanda Fujikoshi Bldg.

23-1 Higashi Gotanda 5-Chome

Tokyo 141, Japan

Tel: +81 3 542 00431 – Fax: +81 3 542 00438

SAES Getters (Nanjing) Co., Ltd.

56 Xingangdadao, Xinshengwei

Nanjing Economic & Technical Development Zone

Nanjing 210038, Jiangsu Province, P.R. of China

Tel: +86 25 8580 2335 – Fax: +86 25 8580 1639

SAES Getters Korea Corporation

7th Fl. Dongwon Bldg. 143-28

Samsung-dong, Gangnam-gu

Seoul 135-877, Korea

Tel: +82 2 3404 2400 – Fax: +82 2 3452 4510/11

SAES Getters S.p.A. – Taiwan Branch Office

6F-1, No. 1071, Zhongzheng Road,

Taoyuan City, Taoyuan County 330

Taiwan, R.O.C.

Tel: +886 3 346 3866 – Fax: +886 3 346 8290

North and South America

SAES Getters USA, Inc.

1122 East Cheyenne Mountain Blvd.

Colorado Springs, CO 80906, USA

Tel: +1 719 576 3200 – Fax: +1 719 576 5025

Disposal

The instrument's disposal must be carried out in compliance with the user's country applicable regulations.

These instructions refer to NEXTorr Z pumps at the end of their normal service life; in case of accidents, fires, flooding or other unforeseeable events, these instructions must be re-evaluated.

This document does not take into account any possible contamination by harmful gases used during service life, which can be present both as oxides in the getter disks or as residuals in pores, gaps, and internal parts of the pump.

All the components of NEXTorr Z are compliant with EU ROHS regulations and equivalent regulations worldwide.

ZAO-based getters are classified as flammable solids according to EC Regulation 1272/2008. SAES recommends disassembling and disposing them as dangerous waste. Special care must be adopted in order to avoid powder generation. Please refer to the *Safety Data Sheet* for further details. Special instruction to handle significant amount of getter material are available on demand.

The ION element is classified as EEE according to the most common worldwide regulations (EU, USA, China, Japan, Korea) and must be disposed according to the local regulations.

Do not dispose of the packing materials in an unauthorised manner. The packaging materials (carton and polyurethane foam) are 100% recyclable.

Warranty conditions

SAES guarantees that the Products delivered shall be free from operational and material defects and shall comply with the construction and functional data and specifications indicated in the Contractual Documents.

This warranty shall have a term of TWELVE (12) MONTHS. For Products which require installation at BUYER's facility by SAES personnel, the warranty shall have a term of TWELVE (12) MONTHS from the date of installation or FOURTEEN (14) MONTHS from the date of delivery, whichever term is shorter. Subject to the remainder of this Article 14, any action by BUYER for any alleged breach of this warranty shall be brought in writing by BUYER within thirty (30) days of BUYER's discovery of the breach. This warranty shall only apply to the BUYER and may not be assigned.

During the term of the warranty set forth above, SAES will promptly repair the Products which for their features can be repaired and which do not conform to the specifications and which BUYER returns to SAES at the address provided. Unless otherwise agreed and specified, BUYER shall be responsible for all transportation charges incurred in returning Products to SAES for repair; BUYER shall have obtained a Returned Material Authorization ("RMA") number and specific shipping instructions from SAES prior to its shipping of the Products to SAES. SAES shall not unreasonably deny BUYER authorization to ship Products to SAES. SAES shall return repaired Products to BUYER, with transportation charges prepaid by SAES, unless otherwise agreed. Additional information is available on the General conditions of sales.



SAES Getters S.p.A. – Italy

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